

Inventory Management System (for Store or Laboratory)

1.Introduction

1. The Inventory Management System plays a vital role in ensuring that materials, supplies, and goods are efficiently tracked and maintained within an organization.
2. During our visit to Kamat Tourist, we studied how the company manages its stock of packaging materials, travel items, and other supplies that are required for their operations.
3. The main focus of this study is to analyze how inventory is recorded, issued, and updated in their system.
4. We also compared the manual methods of maintaining stock (like registers or spreadsheets) with the automated tracking systems (software-based management).
5. The observation highlighted several challenges such as:
 - Overstocking of certain materials.
 - Missing or untracked items. ○ Delays in updating stock records.
 - Human errors during manual entries.
6. To overcome these challenges, we aim to design and propose an automated Inventory Management System that can maintain accurate records, reduce wastage, and ensure timely updates of stock information.
7. This report includes the following sections:
 - Requirement Elicitation – understanding the needs of users and the system.
 - Data Modeling – representing data flow and relationships between stock entities.
 - Input/Output Design – planning user interfaces and report formats.
 - Implementation Plan – describing the steps to develop and deploy the system.

2.Objective of the Survey

- To study and understand the **current inventory management process** followed at *Kamat Tourist*.
- To **analyze how stock items are recorded, issued, and updated** during daily operations.
- To **identify the differences** between manual inventory tracking and any existing automated methods.
- To observe and document **common issues** such as overstocking, missing items, or delayed stock updates.
- To understand the **user requirements** of staff members who manage inventory records.
- To evaluate the **accuracy, efficiency, and reliability** of the current inventory tracking system.
- To **collect information and feedback** that will help design an improved, automated Inventory Management System.
- To ensure that the proposed system can help **reduce human error**, improve stock visibility, and support better decision-making.
- To explore how the same inventory model can be applied to **store or laboratory environments** with minimal modification.

3. Methodology

The methodology followed for the study of the *Inventory Management System* at Kamat Tourist involved a structured approach consisting of field observation, limited data collection, and system analysis. The steps are described below:

1. Field Visit and Preliminary Observation

- A field visit was conducted to Kamat Tourist to gain a general understanding of how inventory operations are managed.
- Due to limited access and staff availability, the observations were primarily visual and process-oriented, focusing on how materials were stored, recorded, and dispatched.
- Photographs and brief notes were taken to document the workflow and the arrangement of stock materials.

2. Secondary Data Collection

- Since detailed internal records were not accessible, secondary information was gathered from general references, previous studies, and standard inventory management procedures.
- Publicly available data and online resources were used to understand common practices in small and medium-scale inventory systems.
- The collected information was used to model an idealized version of how inventory could be managed in similar organizations.

3. Process Analysis

- Based on observations and reference data, the existing system at Kamat Tourist was analyzed to identify key operational activities such as stock entry, issue, and dispatch.
- The current system appeared to rely on manual tracking methods, which may lead to delays, inconsistencies, and record duplication.
- Standard inventory principles were compared against the observed process to highlight potential improvement areas.

4. Problem Identification

- The study identified several limitations in manual inventory handling, including:
 - Lack of real-time updates and traceability.
 - Increased chances of calculation errors and missing entries.
 - Difficulty in monitoring stock movement and generating reports.

- These findings emphasize the need for a computerized Inventory Management System to improve efficiency and data accuracy.

5. Requirement Identification

- General user requirements were formulated based on standard practices and assumed system needs:
 - Functional Requirements: Stock entry, issue tracking, supplier management, report generation.
 - Non-Functional Requirements: Accuracy, ease of use, data security, and reliability.
- These requirements guided the system's conceptual design.

6. System Design and Modeling

- Conceptual models such as Data Flow Diagrams (DFD) and Entity-Relationship Diagrams (ERD) were created to represent data movement and relationships.
- Sample input and output screens were designed for efficient user interaction and report visualization.
- A relational database structure was proposed to ensure smooth data storage and retrieval.

7. Implementation and Evaluation Plan

- A step-by-step plan was proposed for system development, testing, and deployment.
- Technologies such as MySQL (for database) and Python/PHP (for front-end interface) were suggested for implementation.
- Expected outcomes include faster data entry, automated updates, and accurate reporting compared to the current manual approach.

System Design and Modeling:

- Prepared **Data Flow Diagrams (DFD)** and **Entity-Relationship Diagrams (ERD)** to represent system components.
- Designed input forms, output reports, and database structures for the proposed system.

Implementation Planning:

- Planned the steps required to develop, test, and deploy the automated Inventory Management System.
- Suggested suitable technologies such as databases and user interfaces for easy access and operation.

- **Result Evaluation:**

- Compared the proposed system's efficiency with the current manual process.
- Evaluated expected improvements in accuracy, speed, and overall management.

4.SDLC Phases

4.1 Requirement Analysis

1. In this phase, the needs of the users and the organization were studied in detail.
2. Discussions and observations at *Kamat Tourist* helped identify how inventory is currently recorded, issued, and updated.
3. The main requirements identified include:
 - Recording of incoming and outgoing stock.
 - Maintaining supplier and product details.
 - Generating reports on stock levels and usage.
 - Alerts for low stock or overstock situations.
4. Both **functional** (what the system should do) and **non-functional** (performance, reliability, usability) requirements were documented.

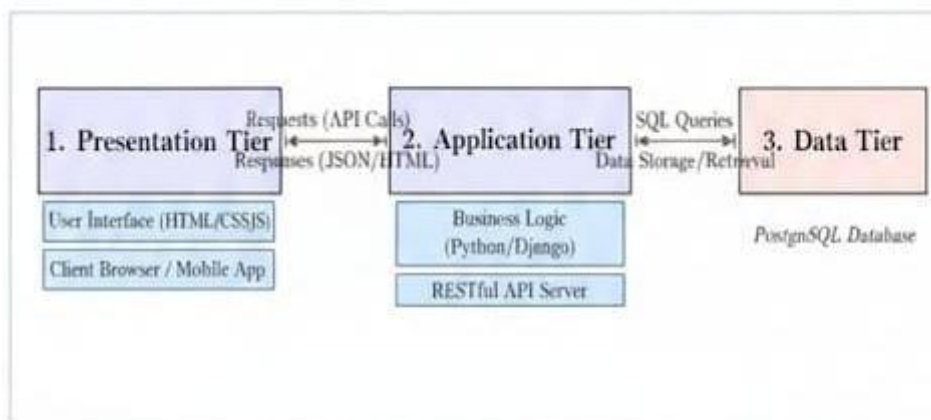
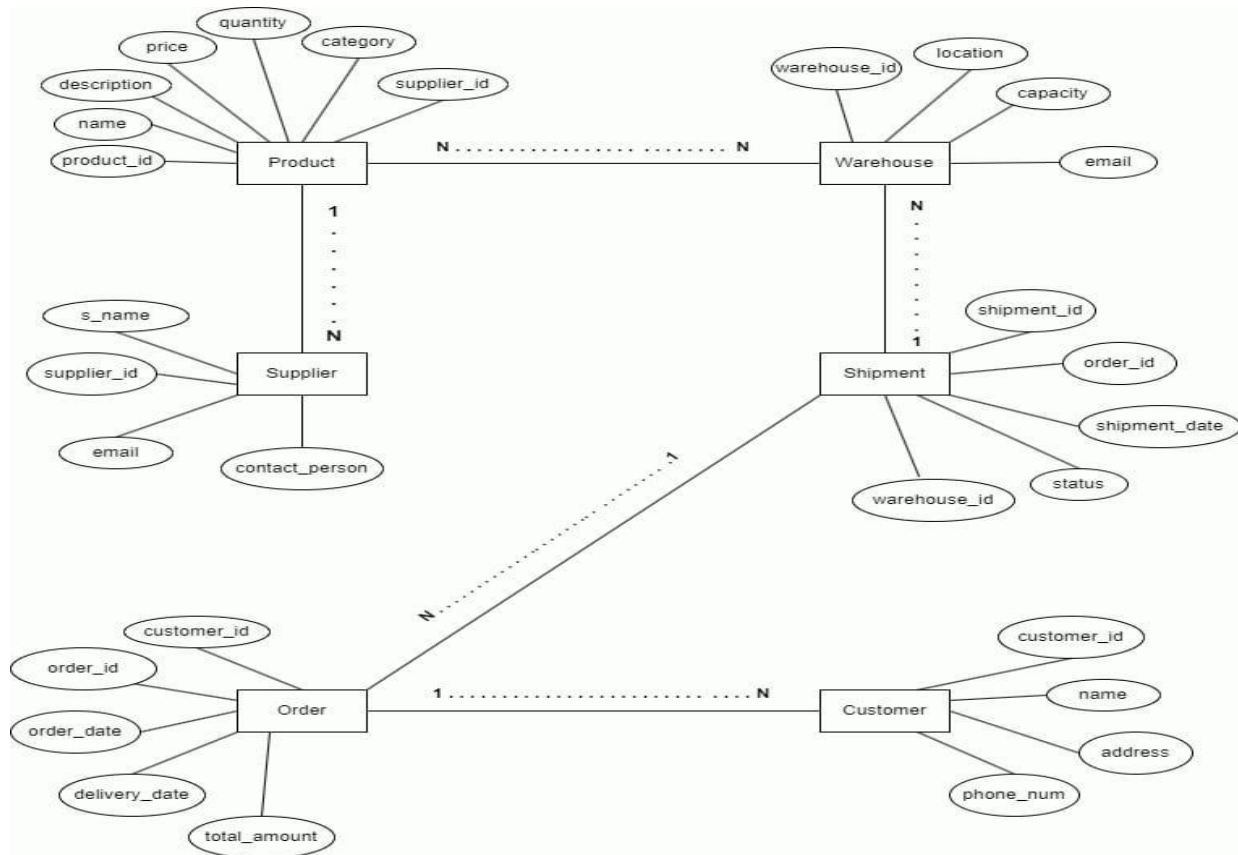
The screenshot displays a web application for bus booking. The top navigation bar includes a home icon, a 'Branch' dropdown, and a 'PNR No' search field. The main section is titled '> Booking' and contains a search form with 'From City' (Bengaluru), 'To City' (Bijapur), and 'Journey Date' (28-10-2025). Below the search form, there are buttons for navigation and a 'Total Routes' button showing 0. At the bottom, there is a table header with columns: Route Name, BusType, Departure, Duration, Arrival, Fare, and View.

4.2 System Design

1. Based on the collected requirements, the system architecture was designed.
2. The **Data Flow Diagram (DFD)** and **Entity Relationship Diagram (ERD)** were created to show how data moves within the system.
3. **Input and Output designs** were planned for stock entry forms, issue forms, and stock reports.
4. The database design includes tables for items, suppliers, stock transactions, and users.
5. The system design ensures modularity and scalability for future enhancements.

Key Entity	Primary Attributes (PK/Other)	Key Relationships	Purpose/Role in System
ITEM	itemID (PK), name, quantityInStock, unitPrice	IS_STORED_AT (to LOCATION), DETAILS_FOR (to ORDER_ITEM), LOGS (to TRANSACTION)	Defines the specific product/material being tracked. Contains current stock levels and base pricing/details.
LOCATION	locationID (PK), name, type	IS_STORED_AT (to ITEM)	Specifies where an item is physically kept (e.g., shelf, freezer, storage bin).
SUPPLIER	supplierID (PK), name, contactPerson, phone	PLACED_BY (to ORDER)	Identifies the external vendor from whom items are procured.
ORDER	orderID (PK), orderDate, status, totalAmount	PLACED_BY (to SUPPLIER), CONTAINS (to ORDER_ITEM)	Records a purchase request or receipt made to a supplier.

Key Entity	Primary Attributes (PK/Other)	Key Relationships	Purpose/Role in System
ORDER_ITEM	orderID, itemID (Composite PK)	DETAILS_FOR (to ITEM), CONTAINS (from ORDER)	Links items to a specific order and records the quantity/cost at the time of the order.
TRANSACTION	transactionID (PK), transactionDate, transactionType, quantity	LOGS (to ITEM), PERFORMS (to USER)	Tracks every movement (in or out) of an item, crucial for auditing and stock adjustment.
USER	userID (PK), username, role	PERFORMS (to TRANSACTION)	Identifies the personnel responsible for system actions like inventory adjustments or order processing.



Inventory Management System

Data flow (narrative)

1. STORE STAFF/MANAGER Interaction

- Input to IMS: Sends New Product Data / Updates (for item setup), Stock Adjustment Requests, and Report Requests (e.g., Low Stock Report, Usage Report).
- Output from IMS: Receives Status Updates / Stock Levels, Inventory Reports, and Reorder Lists.

2. SUPPLIER Interaction:

- Input to IMS: Sends Shipment Confirmation (for incoming goods) and Invoice/Billing Data.
- Output from IMS: Receives Purchase Orders (based on reorder calculations).

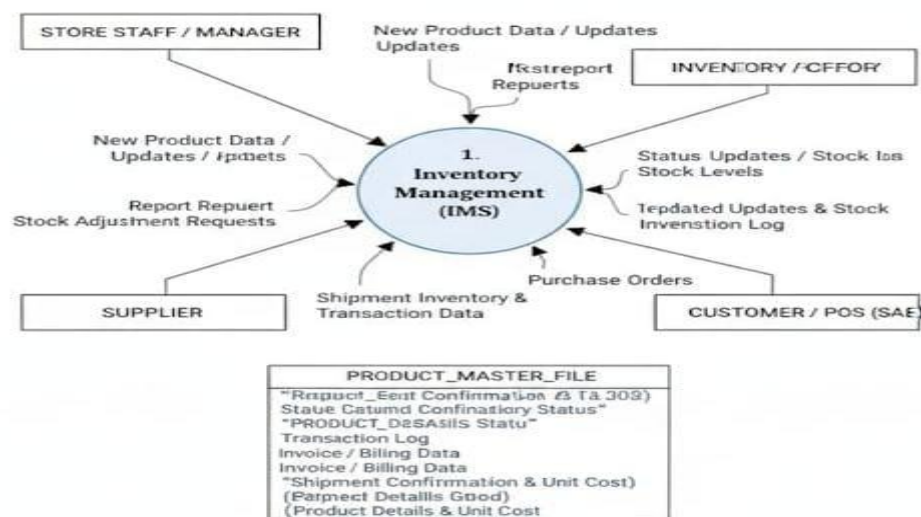
3. CUSTOMER/POS Interaction:

- Input to IMS: Sends Sales Transaction Data / Checkout (records item quantity sold).
- Output from IMS: Receives Item Price & Stock Lookup (for sales floor verification).

4. Data Store Interaction:

- The IMS interacts with the PRODUCT_MASTER_FILE to retrieve Product Details & Unit Cost for sales processing and reporting.
- The IMS sends Updated Inventory Status and Transaction Log data to the data store.

Data Flow Diagram: Management System



4.3 Implementation

1. In this phase, the actual coding and system development take place.
2. A suitable programming language and database (e.g., Python with MySQL or PHP with SQL) can be used for the implementation.

3. The front-end design includes user-friendly forms for adding, viewing, and updating stock records.
4. The system automates tasks like stock updates, issuing reports, and generating alerts.
5. Role-based access can be implemented for admin and staff users to ensure data security.

4.4 Testing

1. The developed system is tested to ensure it functions correctly and meets user requirements.
2. **Unit Testing** – individual modules are tested (e.g., stock entry, issue form).
3. **Integration Testing** – verifies data flow between modules.
4. **System Testing** – ensures all parts work together as expected.
5. **User Acceptance Testing (UAT)** – staff at *Kamat Tourist* or similar organizations use the system and provide feedback.

4.5 Deployment

1. After successful testing, the system is deployed for actual use.
2. The application is installed on the organization's computers or hosted online for easy access.
3. Data from old records (manual or Excel) can be imported into the new system.
4. Staff members are trained on how to use the software efficiently.
5. Backup and data recovery measures are implemented before final deployment.

4.6 Maintenance

1. After deployment, continuous maintenance ensures smooth system performance.
2. Regular updates are made to fix bugs, improve performance, and add new features.
3. Periodic database cleanup and backup are carried out to prevent data loss.
4. User feedback is collected to improve usability and adapt to changing requirements.
5. The system is monitored to ensure long-term reliability and security.

5.Observations & Findings

- The inventory at *Kamat Tourist* is primarily used for managing packaging materials, travel items, and related supplies for dispatching packages.
- Most of the inventory processes are **handled manually** using registers or basic spreadsheets, which increases the chances of human error.
- **Stock updates are not done in real-time** — this often leads to delays in knowing the exact availability of items.
- There is **no centralized database** to store stock information, resulting in difficulties in tracking issued and remaining items.
- During observation, it was noticed that **duplicate entries and calculation mistakes** occur due to manual recordkeeping.
- **Overstocking and understocking** problems were identified, mainly due to lack of automated alerts or reorder level notifications.
- The **process of issuing stock** involves multiple approvals, which sometimes causes delays in operations.
- The **manual verification of stock** at the end of each month is time-consuming and prone to inconsistencies.
- Employees expressed a need for an **automated system** that could simplify daily tasks such as data entry, report generation, and inventory tracking.
- It was found that there is **no proper report generation system**, making it hard for management to analyze stock usage or trends.
- Overall, the observation showed that implementing a **computerized Inventory Management System** would improve accuracy, reduce workload, and enhance operational efficiency.

6.Geotagged Group Photo



7. Conclusion

1. The study of the existing inventory process at *Kamat Tourist* provided valuable insights into how stock management is currently carried out.
2. It was observed that most inventory-related tasks are performed **manually**, which leads to common issues such as **delayed updates, data inaccuracies, and time-consuming verification**.
3. The lack of an **automated tracking system** makes it difficult to monitor real-time stock levels, causing problems like **overstocking and stock shortages**.
4. Implementing an **automated Inventory Management System** would help overcome these challenges by ensuring accurate recordkeeping, faster data entry, and instant report generation.
5. The proposed system will also enable **better decision-making** by providing updated information on stock levels, usage patterns, and reorder requirements.
6. Automation will significantly **reduce human errors**, save time, and improve coordination among staff members involved in stock handling.
7. Overall, this project highlights the importance of a well-structured, computerized inventory system for organizations like *Kamat Tourist* to improve **efficiency, reliability, and productivity**.

8. Outcomes and Learning

- Gained a clear understanding of how **inventory management** functions in a real-world organization like *Kamat Tourist*.
- Learned to **analyze and compare manual and automated systems** for stock management.
- Understood the importance of maintaining **accurate, up-to-date stock records** to prevent overstocking and shortages.
- Identified common **issues faced in manual systems**, such as data duplication, human error, and delayed updates.
- Developed skills in **requirement elicitation** — understanding user needs and system expectations through observation and interaction.
- Learned to design **data models (DFD and ER diagrams)** for representing the logical flow of data in an inventory system.

- Understood how **SDLC phases** (Requirement Analysis, Design, Implementation, Testing, Deployment, and Maintenance) are applied in a real project scenario.
- Improved knowledge of **system design and interface development** for creating efficient input/output screens and reports.
- Recognized the benefits of **automation and database management** in improving operational efficiency and accuracy.
- Gained practical experience in preparing a **structured project report**, covering analysis, findings, and proposed system solutions.
- Learned the importance of **collaboration and communication** while gathering data from field visits and interacting with staff.
- Overall, the study enhanced both **technical and analytical skills**, contributing to better understanding of software development in real-life business environments.

9. References

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